

Muhammad Abdullah

Computer Engineering Student | AI · Cloud · Backend Development

Lahore, Pakistan | whello544@gmail.com | +92 314 1679454 | LinkedIn | GitHub | Portfolio | HuggingFace

EDUCATION

BS Computer Engineering — COMSATS University Islamabad, Lahore Campus	Sep 2023 – Jun 2027
Relevant Coursework: VLSI Design, Digital System Design, Data Communication & Networks, Artificial Intelligence, Machine Learning	
Intermediate (ICS) — Grade A — Punjab Group of Colleges	2021 – 2023

EXPERIENCE

Intern Web Developer — MKY Global Forwarding (Remote)	Mar – Jul 2025
- Deployed a full logistics website for a startup, improving operational efficiency across key workflows. - Collaborated with a development team on real-world projects using HTML, CSS, JavaScript, and SQL.	
C++ Developer Intern — Rhombix Technologies (Remote)	Jan – Mar 2025
- Built efficient, scalable C++ applications focused on Data Structures, Algorithms, and OOP principles. - Delivered a feature-complete To-Do List manager and an Encryption Tool (Caesar cipher).	

PROJECTS

Bank Reconciliation API	2026
<i>FastAPI · Pandas · PostgreSQL · REST API</i>	
Automated bank statement reconciliation API with fuzzy matching. Upload two CSVs, get matched/unmatched transactions + balance difference in seconds.	
Data Normalizer API	2026
<i>Python · FastAPI · Pandas · Async</i>	
Async API to upload dirty files, auto-detect schemas, normalize column names, handle nulls, and download clean data.	
FastAPI JWT Auth API	2026
<i>FastAPI · JWT · Security · REST</i>	
Secure REST API with JWT authentication, role-based access and token refresh — production-style auth flows.	
AEGIS Swarm — Multi-Agent AI System	2026
<i>Python · HuggingFace Spaces · Fine-tuned LLM · Vision-Language Model</i>	
- Designed an autonomous multi-agent swarm for executing complex tasks with collaborative agent communication layers. - Fine-tuned a dedicated AEGIS-SWARM LLM and a multimodal Visual Agent for agentic coordination; both deployed on HuggingFace.	
8×8 Sparsity-Aware Systolic Array — CEP Research Project	2026
<i>VHDL · Verilog · RTL Simulation · Hardware Design</i>	
- Behavioural RTL simulation of an 8×8 sparsity-aware systolic array for efficient neural network matrix inference. - Dynamic power gating detects zero-valued elements in real time, gating clock/power domains to reduce energy consumption.	
Fake News Detector — Hackathon Winner	2025
<i>Python · NLP · Text Classification · Gradio</i>	
AI-driven misinformation detection tool built during a hackathon; uses NLP classification to flag fake news in real time.	
FlumenIntel App & News Analyzer App	2025
<i>Python · Gradio · Groq · NewsAPI · HuggingFace Spaces · SQLite</i>	
- FlumenIntel: real-time data intelligence app with ML-powered visualisation, deployed on HuggingFace Spaces. - News Analyzer: AI news app featuring sentiment analysis, LLM reasoning, and credibility scoring with persistent article history.	

CERTIFICATIONS & BADGES

AWS SimuLearn: Serverless Developer — Amazon Web Services (2026) — 24h hands-on labs on live AWS console: Lambda, API Gateway, DynamoDB, event-driven architecture & CI/CD. AWS Cloud Quest: Generative AI Practitioner — Amazon Web Services (2026) — GenAI solutions on AWS AI services: prompt engineering, chatbots, content moderation & fine-tuning. AWS AI & ML Scholars — Udacity (2026) — Passed project: "Using a Pre-trained Image Classifier to Identify Dog Breeds". - Generative AI for Everyone — DeepLearning.AI / Coursera (2025) - Create Generative AI Apps on Google Cloud — Google Cloud Skills Boost (2025) - Generative AI Certification — Pak Angels (2025) - Programming for Everybody & Python Data Structures — University of Michigan / Coursera (2025) - Google Cloud Challenge Completion Badge — Qwiklabs (2026) - DataCamp Track: SQL, Python for Developers, APIs in Python, Importing Data, Data Engineering, Cloud Computing (2026)	
---	--

VOLUNTEERING & LEADERSHIP

Microsoft Learn Student Ambassador (MLSA) — COMSATS Lahore	Feb 2025 – Jun 2025
Member & Co-Strategist: led technical workshops on AI and cloud, coordinated knowledge-sharing initiatives across campus.	

IEEE — COMSATS Lahore	Feb 2025 – Sep 2025
Active member contributing to engineering society events, technical workshops, and student outreach programmes.	
Youthe — COMSATS Lahore	Feb 2025 – Present
Engaged in student-led development events and community service initiatives.	

SKILLS

- Languages:** Python, C++, Java, JavaScript / TypeScript, SQL, HTML/CSS
- AI & ML:** LLMs, Generative AI, YOLOv8, TensorFlow / TFLite, HuggingFace, Gradio, RAG, pgvector, Claude API
- Hardware:** Verilog, VHDL, FPGA (Xilinx Vivado), GNU Radio, RTL-SDR, VLSI Design, Proteus, PCB Design
- Web & Cloud:** Next.js, FastAPI, Supabase, Redis, Vercel, Railway, Google Cloud, Fly.io, AWS (Lambda, API Gateway, S3, DynamoDB, CloudWatch)
- Tools:** Git / GitHub, Cisco Packet Tracer, MATLAB, AutoCAD, Figma, Cursor
- Soft Skills:** Technical leadership, problem-solving, teamwork, public speaking / workshops